Nitrites

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Identity Sheet

Property	Value
Name	Nitrite
Other names	Nitrite ion, NO_2^-
Chemical formula	NO_2^-
Molecular weight	46.01 g/mol (as ion)
CAS number	14797-65-0 (nitrite ion)
PubChem CID	946
SMILES	[N-]=O (nitrite ion)
Appearance	Colourless in solution

Structural formula:

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O=N-O(-) (bent geometry, bond angle  115 deg)
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Nitrite (NO_2^-) is an inorganic anion formed by bacterial reduction of nitrate (NO_3^-) in urine. It is not a metabolic product of human cells but rather a **marker of bacterial presence** in the urinary tract. In the urinary context, nitrite is exclusively a surrogate biomarker for bacteriuria. Nitrate is normally present in urine from dietary sources and is reduced to nitrite by Gram-negative bacteria possessing nitrate reductase (e.g., *E. coli*, *Klebsiella*, *Proteus*).

Molecules Not to Be Confused With

Molecule	Formula	MW (g/mol)	Key Difference
Nitrate (NO_3^-)	NO_3^-	62.00	Precursor of nitrite; normally present in urine from diet; not indicative of infection
Nitric oxide (NO)	NO	30.01	Gaseous signalling molecule; not stable in aqueous solution
Nitrogen dioxide (NO_2)	NO_2	46.01	Toxic gas (neutral molecule, not the ion); environmental pollutant

Medical Information

Origin

Endogenous Nitrite is **not endogenously produced** in the urinary tract. Its presence in urine results from bacterial enzymatic activity: - **Nitrate reductase** in Gram-negative bacteria (primarily Enterobacteriaceae: *E. coli*, *Kleb-*

siella, *Proteus*, *Citrobacter*, *Enterobacter*) reduces dietary urinary nitrate (NO_3^-) to nitrite (NO_2^-). - This reaction requires a **bladder dwell time of >= 4 hours** for sufficient nitrite accumulation.

Physiologically, small amounts of nitrite can be formed systemically from nitric oxide (NO) metabolism and dietary nitrate reduction by oral bacteria, but this does not contribute meaningfully to urinary nitrite.

Exogenous

- **Dietary nitrate** is the ultimate source: green leafy vegetables (spinach, lettuce, beetroot), cured meats. Nitrate is absorbed, enters the blood, and is excreted in urine where bacteria may convert it.
- Direct dietary nitrite intake (from cured meats, preservatives E249/E250) is absorbed but does not significantly affect urinary nitrite levels.

Primary & Secondary Biological Roles

Primary role: - **UTI diagnostic marker:** Nitrite has no physiological role in urine. Its presence is an indirect indicator of bacteriuria, specifically Gram-negative organisms with nitrate reductase activity.

Secondary roles: - **Systemic nitrate-nitrite-NO pathway:** In blood, nitrite serves as a reservoir for NO production (vasodilation) under hypoxic conditions. This pathway is unrelated to urinary nitrite testing.

Catabolism and Elimination Pathway

- Nitrate is freely filtered at the glomerulus and partially reabsorbed.
- ~60–70% of dietary nitrate is excreted in urine as intact nitrate.
- Nitrite (when produced by bacteria in urine) is unstable and can be further reduced to N_2, NH_3, or NO by bacterial enzymes.
- In acidic urine, nitrite decomposes to nitrous acid (HNO_2) and NO gas.

Expression in Humans

Normal Levels

Compartment	Reference Range
Urine nitrite	Absent (negative on dipstick)
Urine nitrate	40–200 mg/day (diet-dependent)
Blood nitrite	~0.1–0.5 µmol/L (from NO metabolism, not related to UTI)

Factors Influencing Levels **Positive urinary nitrite:** - UTI with Gram-negative bacteria (E. coli, Klebsiella, Proteus) - Sufficient dietary nitrate intake - Adequate bladder dwell time (>=4 h, typically overnight) - Urine pH <6.0 (some bacteria more active)

Negative urinary nitrite (despite infection): - Gram-positive organisms (Enterococcus, Staphylococcus) — lack nitrate reductase - Short bladder dwell time (<4 h, frequent voiding) - Low dietary nitrate intake - Very dilute urine - Ascorbic acid interference (false negative) - Yeast UTI (Candida)

Associated Pathologies

Condition	Nitrite Status	Key Symptoms
Uncomplicated UTI (cystitis)	Positive (~45–60% sensitivity)	Dysuria, frequency, urgency
Pyelonephritis	Often positive	Fever, flank pain, nausea

Condition	Nitrite Status	Key Symptoms
Asymptomatic bacteriuria	May be positive	No symptoms; screening in pregnancy
Gram-positive UTI	Negative despite infection	Varies by organism

Presence in Urine

Should it be normally present? No — nitrite is absent in uninfected urine. Its presence always indicates bacterial conversion of nitrate.

Normal urinary levels: Undetectable (negative).

Form in urine: Nitrite ion (NO_2^-) — dissolved, fully ionic. At low pH, exists partly as nitrous acid (HNO_2, pKa 3.3), but at typical urine pH 5–7, predominantly ionic.

Pathological significance:

Urinary Nitrite	Possible Causes	Prevalence
Positive	Gram-negative bacteriuria (UTI or ASB)	UTI: ~150 million cases/year globally
Negative	Normal; or Gram-positive UTI; or insufficient dwell time	—

Solubility: Nitrite salts are highly soluble in water. No crystallisation issues.

Detection in Urine

Available Clinical Assays

1. **Dipstick nitrite test (Griess reaction, gold standard screening):**
   - **Principle:** Nitrite in acidic medium reacts with aromatic amine (p-arsanilic acid or sulfanilamide) to form a diazonium salt, which couples with a naphthylamine derivative (N-(1-naphthyl)ethylenediamine or tetrahydrobenzoquinoline) → pink-red azo dye.
   - **Detection:** Visual colour change on reagent pad; graded negative/positive (some strips: trace/+/++).
   - **LOD:** ~0.05–0.1 mg/dL nitrite (~11–22 μmol/L).
   - **Sensitivity for UTI:** 45–60%. **Specificity:** 85–98%.
   - **Absorption of product:** ~540 nm (azo dye).
   - **Advantages:** Rapid (<60 s), inexpensive, highly specific when positive.
   - **Disadvantages:** Low sensitivity (misses Gram+ organisms, short dwell time); semi-quantitative.
2. **Colorimetric Griess assay (quantitative, laboratory):**
   - **Principle:** Same Griess reaction but performed in solution with spectrophotometric readout.
   - **Detection:** Absorbance at **540–548 nm**.
   - **LOD:** ~0.01 mg/dL (~2 μmol/L).
   - **Advantages:** Quantitative, more sensitive than dipstick.
   - **Disadvantages:** Requires spectrophotometer; not point-of-care.
3. **Chemiluminescence (NO detection after reduction):**
   - **Principle:** Nitrite reduced to NO by acidified triiodide or vanadium(III); NO detected by chemiluminescent reaction with ozone.
   - **Detection:** Photon emission at ~600–900 nm.

- **LOD:** ~1 nM (0.05 µg/L).
- **Advantages:** Ultra-sensitive, gold standard for low-level nitrite.
- **Disadvantages:** Complex instrument (Sievers NOA), research use only.

Optimal Urine Type for Measurement

- **First morning urine (first void)** is strongly recommended — longest bladder dwell time (overnight, >=4 h), maximising nitrate-to-nitrite conversion.
- Random specimens collected after recent voiding may be false negative.
- **No preservatives** needed — test immediately or within 2 h (nitrite is unstable, decomposes in acidic conditions and with bacterial overgrowth).
- Refrigerate if delay unavoidable.

Actual Gold Standard

The **urine dipstick Griess reaction** is the universal screening method for nitrite. For quantitative nitrite measurement, the **solution-phase Griess assay** with spectrophotometric readout at 540 nm is the laboratory reference. For trace-level research, **chemiluminescence-based NO analysers** (Sievers NOA 280i) after reductive conversion provide the highest sensitivity.

Interferences in Measurement

Interference	Effect	Mechanism
Ascorbic acid (vitamin C)	False negative	Reduces diazonium salt, preventing azo dye formation; significant at >25 mg/dL
High specific gravity	Variable	Concentrated urine may delay colour development
Bacterial overgrowth (delay)	False negative	Further reduction of nitrite to N_2 or NH_3
Low dietary nitrate	False negative	Insufficient substrate for bacterial conversion
Acidic urine pH (<5)	Nitrite decomposition	$HNO_2 \rightarrow NO + H_2O$ (spontaneous decomposition)
Gram-positive organisms	False negative	Lack nitrate reductase (no nitrite produced)
Urobilinogen (high)	Potential colour interference	Orange/brown may mask pink colour (rare)
Phenazopyridine	Colour interference	Orange dye obscures colour interpretation

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- **UV-Vis (Griess product):** Azo dye absorbance at **540–548 nm**. LOD: ~2 µmol/L (Griess assay in solution).
- **UV-Vis (direct nitrite absorption):** Nitrite absorbs weakly at **354 nm** (ε ~23 $M^{-1}cm^{-1}$) and **210 nm** (ε ~5,300 $M^{-1}cm^{-1}$). Direct UV detection at 210 nm gives LOD ~10 µmol/L, but severe urine matrix interference.
- **NIR:** No useful absorption. Not applicable.
- **Sample prep (Griess):** Add Griess reagent (sulfanilamide + NED in phosphoric acid); 10 min incubation at RT.

Fluorescence Detection

- Nitrite itself is **non-fluorescent**. Indirect approaches:
- **DAN (2,3-diaminonaphthalene) assay:** Nitrite reacts with DAN in acidic conditions to form fluorescent 2,3-naphthotriazole (NAT). **Ex 365 nm / Em 410 nm**.
  - LOD: ~10–50 nM (0.5–2 μg/L) — 100x more sensitive than Griess.
  - Sample prep: Acidification + DAN reagent + 10 min incubation + NaOH to stop reaction.
- **Fluorescent probes (rhodamine-based):** Cu^{2+}-mediated catalytic diazotisation with fluorescent amine probes. Ex 520 / Em 580 nm. LOD: ~100 nM.
- **Quantum dot quenching:** CdTe QDs quenched by nitrite-derived NO. Ex 350 / Em 530 nm. LOD: ~50 nM.

Raman Detection Nitrite has limited Raman features:

Peak (cm^{-1})	Assignment
~1330	NO_2 symmetric stretch (νs)
~815	N-O bend (δ)
~1240	NO_2 asymmetric stretch (νas)

- Conventional Raman: LOD ~10 mM (far too high for clinical use).
- **SERS:** Silver nanoparticle substrates with 532/633 nm excitation. LOD improved to ~1–10 μmol/L. Still marginal for clinical nitrite detection.
- **Sample prep:** Direct measurement or mixing with colloidal nanoparticles.

FTIR Detection

Band (cm^{-1})	Assignment	Notes
~1275	NO_2 asymmetric stretch	—
~1330	NO_2 symmetric stretch	Strongest band
~830	N-O bending	—

- ATR-FTIR detection of nitrite in urine not practical — nitrite levels (when present) are typically <0.5 mmol/L, below FTIR sensitivity in complex urine matrix.
- **Research gap:** No validated FTIR method for urinary nitrite.

Voltammetry Detection

- **Direct oxidation:** Nitrite is electrochemically oxidised to nitrate at modified electrodes: $NO_2^- \rightarrow NO_3^- + 2H^+ + 2e^-$. Oxidation peak at **+0.7 to +0.9 V vs Ag/AgCl** on glassy carbon, lower on modified electrodes.
- **Electrode materials:** Carbon nanotubes, graphene oxide, gold nanoparticles, metal-organic frameworks (MOFs) reduce overpotential and improve selectivity.
  - LOD: ~0.1–1 μmol/L on nanostructured electrodes.
  - Linear range: 1–1000 μmol/L.
- **Amperometric sensors:** Applied +0.8 V; continuous monitoring.
  - LOD: ~0.5 μmol/L.
- **Sample prep:** Dilution in buffer (pH 4–7); minimal.
- Selectivity concern: ascorbic acid, uric acid oxidise at similar potentials.

Other Detection Technologies

1. **Ion chromatography (IC):** Anion exchange column with conductivity detection. LOD: ~0.1 µmol/L. Sample prep: filtration, dilution. Gold standard for accurate quantification in research.
2. **Capillary electrophoresis:** UV detection at 210 nm. LOD: ~1 µmol/L. Fast separation.
3. **Electrochemical paper-based sensors:** Screen-printed carbon electrodes on paper with DPV. LOD: ~1–5 µmol/L. Low cost, disposable.
4. **Smartphone-based Griess colorimetry:** Paper strip + phone camera + colour analysis app. LOD: ~5–10 µmol/L. Point-of-care potential.

Summary Table: Detection Methods Comparison

Method	LOD	Wavelength / Key Parameter	Sample Prep	Strengths	Limitations
Dipstick Griess (screening)	~11 µmol/L	~540 nm (azo dye, visual)	None	Rapid, cheap, specific	Low sensitivity for UTI
Griess assay (quantitative)	~2 µmol/L	540–548 nm	Reagent addition	Quantitative	Spectrophotometer needed
Chemiluminescence (NOA)	~1 nM	600–900 nm (CL)	Reductive conversion	Ultra-sensitive	Complex, research only
DAN fluorescence	~10–50 nM	Ex 365 / Em 410 nm	DAN + acid + NaOH	Very sensitive	Derivatisation needed
SERS	~1–10 µmol/L	1330 cm^{-1}; Ex 532/633 nm	Nanoparticles	Label-free	Marginal sensitivity
Voltammetry	~0.1–1 µmol/L	+0.7–0.9 V oxidation	Buffer dilution	Sensitive, real-time	Ascorbic acid interference
Ion chromatog-raphy	~0.1 µmol/L	Conductivity	Filtration	Accurate, multi-anion	Instrumentation
Paper electro-chemical	~1–5 µmol/L	DPV at carbon electrode	None	Disposable, low cost	Semi-quantitative

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